Fixed-threshold transfer and equal-review ranking in breast ultrasound segmentation quality control

Running title Threshold transfer and review budgets

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Abstract

Objectives To distinguish transfer of fixed numerical review thresholds from ranking at a fixed review count for breast ultrasound segmentation quality control.

Methods Frozen Segment Anything Model predictions used reference-derived, known-location boxes. Confidence and prompt-disagreement thresholds targeting 20 % review in 213 development patients were transferred to 851 internal and 252 external patients. Low quality was reference Dice <0.70. Equal-review ranking and a post hoc eight-budget analysis assessed policy dependence.

Results Confidence-based detection was 71/93 internally and 10/16 externally: −13.84 percentage points (95 % confidence interval −38.85 to 7.45). Review fractions were 20.45 % and 13.89 %, respectively. Externally, disagreement detected 13/16 low-quality masks at its transferred cutoff versus 10/16 for confidence; at 50 reviews per method, the corresponding counts were 13/16 and 14/16. Across targets from 5 % to 50 %, neither score consistently detected more low-quality masks.

Conclusions Fixed-threshold transfer and equal-review ranking answer different deployment questions. Observed score comparisons depended on the policy and budget; external detection remained imprecise. Reporting detection, review fraction and misses together makes the operating rule explicit.

Keywords domain shift; external testing; prompt perturbation; selective prediction; vision foundation models

Introduction

The Segment Anything Model (SAM) can delineate an object from a spatial prompt, such as a box around a lesion [1]. Medical adaptations of SAM have evaluated box-prompted segmentation across several modalities, including breast ultrasound [2]. When a segmentation is used to support a measurement or other downstream analysis, a separate question is which outputs should be sent for review. A high overall Dice coefficient can coexist with a subset of masks that have poor agreement with a reference contour.

Case-level quality estimation without a new manual mask has been studied with reverse classification accuracy [3]. For promptable models, Kinalioglu combined mask changes under small prompt perturbations with model confidence for failure screening [4]. Loke et al. directly compared prompt-perturbation uncertainty with predicted IoU in SAM/MedSAM; most case-level differences remained unresolved after paired testing and multiplicity correction [5]. These studies motivate a separate operating question: a score can be evaluated by transferring a numerical cutoff or by allocating the same number of reviews. The former preserves a development operating rule; the latter compares rankings at a specified capacity. Their implications for detection and workload need to be distinguished when scores are transferred between datasets.

We evaluated these two operating policies using model-confidence and prompt-disagreement risks on identical initial masks from a frozen SAM. Each score was calibrated to a 20 % development review target in BUS-BRA, then its numerical cutoff was applied unchanged to internal BUS-BRA and external BrEaST patients. The prespecified primary contrast, H1, compared external-minus-internal low-quality detection for confidence-based method A; H2 assessed its review fraction, and H3 compared the two scores on the same low-quality external masks at the locked cutoffs. Equal-review ranking addressed the capacity-constrained question separately. Complete-budget and resampling analyses examined the scope and stability of the comparison, while diagnostic-stratum accounting and prediction contours provided context for workload differences and missed cases.

Materials and methods

Study design and datasets

This retrospective evaluation used anonymized images and manual lesion contours from two public breast ultrasound datasets. Reporting was informed by the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) 2024 update [6]. The patient was the analysis unit, with one image and one primary reference mask per patient. Figure 1 shows the development, internal test and external test cohorts. Table 1 describes their composition.

BUS-BRA version 1.0 contains 1,875 images from 1,064 patients [7, 8]. We used a patient-level split stratified by benign or malignant lesion status to obtain 213 development and 851 internal test patients (seed 20261105). We selected one eligible image per patient using seed 20261106. The two cohorts had disjoint patient identifiers. We verified image and mask pairs and used the actual pixel dimensions when these differed from metadata. An initial 40-case feasibility run used a subset of development patients. The full development set supplied the formal thresholds.

The BrEaST version 1 archive includes 256 scans, each from a different patient [9, 10]. We matched the official image and main tumor mask by case identifier. Four normal cases lacked a primary lesion mask, leaving 252 external test patients. The primary reference was the provided main tumor contour. Fourteen additional masks were outside that reference definition. The audit found zero exact image matches between the datasets.

[Place Table 1 near here]

Prompted segmentation and risk scores

We used the official SAM ViT-B checkpoint (sam_vit_b_01ec64.pth) with frozen float32 weights and single-mask output [1]. The original source commit, checkpoint hash and software versions are recorded in Table S1. We decoded images to red, green and blue (RGB) channels. BrEaST images were alpha-composited over black before conversion to RGB. This changed the RGB pixel values of eight images relative to discarding the alpha channel. Reference masks were read as binary arrays.

We obtained the initial prompt box from the tight bounding box of the nonempty reference mask, expanded each horizontal side by 10 % of the width and each vertical side by 10 % of the height, and clipped the result to the image. This is a reference-informed, known-location (oracle-localization) setting, not a test of autonomous lesion localization. We encoded the image once and used the resulting embedding for the initial prediction and four additional predictions (Figure 2). The fixed 10 % context margin was used consistently across datasets.

Method A calculated risk as one minus the raw predicted intersection over union (IoU) returned for the initial mask. For method B, we shifted the initial clipped box left and right by 5 % of its width and up and down by 5 % of its height, clipping each shifted box to the image. We calculated the Dice coefficient between the initial predicted mask and each of the four shifted-prompt masks. Risk B was one minus the mean of these four Dice coefficients. We retained all four directions when clipping produced coincident boxes. The consistency Dice coefficient was defined as 1 when both predicted masks were empty. The 5 % translations define a deterministic local stability probe relative to box size. Method B adds four mask-decoder passes while reusing the image embedding. Recorded inference times cover the combined pipeline; A-only and incremental B times were not separately benchmarked.

For binary masks, the Dice coefficient is twice the number of shared foreground pixels divided by the sum of foreground pixels in the two masks. The quality reference for both methods was Dice between the initial prediction and the manual mask. An empty initial prediction against a nonempty reference would contribute a Dice coefficient of 0. We defined low quality as reference Dice below 0.70, an operational cutoff for this evaluation.

OpenAI Codex assisted with study-design discussions and implementation of research, statistical and plotting code. The pipeline processed public images with frozen SAM weights, and figures were rendered conventionally from recorded data and captured predictions. The human authors retain responsibility for the implementation and calculated outputs. A precise underlying Codex model identifier was not available in the archived study records.

Development threshold selection

For each method, we searched the distinct development risk scores and the option to flag zero cases. A case was flagged if its score met or exceeded the candidate threshold. The selected threshold gave the review fraction closest to 20 %. Cases tied at the cutoff were flagged together. When two choices were equally close to the target, we selected the one with fewer flagged cases. The exact thresholds were locked before test scoring and applied to both test cohorts. The 20 % target defines a prespecified one-in-five review scenario rather than an empirically measured clinical review capacity.

Outcomes and statistical analysis

The analysis included cases with a valid reference Dice and both finite risk scores. The procedure allowed one retry with the same configuration after a technical failure and recorded persistent failures separately. We report low-quality detection as the number of flagged low-quality masks divided by the number of low-quality masks. Simulated review is the number of flagged cases divided by all scored cases. Misses per 100 patients is 100 times the number of unflagged low-quality masks divided by all scored cases.

The dated analysis plan specified three contrasts. H1, the primary contrast, compared method A low-quality detection in BrEaST minus BUS-BRA internal testing. H2 used the same cohort order for method A simulated review. H3 assessed whether method B detected a larger fraction of the same low-quality BrEaST cases when the two cutoffs gave equal review counts in development. The paired contrast was B minus A. Differences are expressed in percentage points. We calculated two-sided 95 % confidence intervals (CIs) using Newcombe score method 10 for independent proportions in H1 and H2 [11] and its paired counterpart in H3 [12]. H2, H3 and supplementary intervals are nominal, with no multiplicity adjustment.

We repeated detection analyses with reference Dice cutoffs of 0.50 and 0.80 while retaining both locked risk thresholds. We also ranked the external cases by each risk score and selected the top 50 per method, the count closest to a 20 % review fraction among 252 cases. Risk ties followed the unmodified archived patient-identifier strings in lexicographic order (for example, "1", "10", "100"), rather than numeric or zero-padded image-identifier order. This equal-review comparison used a common number of reviews in the observed external cohort; its paired interval is conditional on the observed rankings.

As post hoc descriptive quality-control measures, we additionally tabulated true-positive (TP), false-positive (FP), true-negative (TN) and false-negative (FN) counts, sensitivity (TP/(TP+FN)), specificity (TN/(TN+FP)), positive predictive value (PPV; TP/(TP+FP)) and negative predictive value (NPV; TN/(TN+FN)). Low-quality segmentation was the positive condition and a review flag the positive test. We used Wilson 95 % intervals for these proportions. These measures assess mask-quality screening, not diagnosis of breast cancer.

We additionally performed a post hoc exploratory stratification by dataset-reported benign/malignant classification, retaining the locked risk thresholds and Dice < 0.70 definition. BUS-BRA labels came from the official Pathology field, linked by image and patient identifiers. BrEaST labels came from Classification, linked by CaseID and verified against the selected filenames; 197 diagnoses were biopsy-confirmed and 55 were confirmed by follow-up. We report stratum counts, review fractions and detection

with Wilson 95 % intervals. No stratum-specific recalibration, hypothesis test or interaction test was performed. We also audited all external low-quality cases missed by A, partitioned by the B review flag.

Post hoc policy sensitivity and diagnostic accounting

The following exploratory analyses were specified for this revision after the original study. They retain the original scores, low-quality definition and H1–H3. We examined development review targets of 5 %, 10 %, 15 %, 20 %, 25 %, 30 %, 40 % and 50 %, calibrating a separate threshold for each method with the same tied-group and nearest-target rules. Selection used development risks and counts, without quality labels. Each numerical threshold was transferred unchanged to both test cohorts. A separate equal-review policy selected the nearest integer to the target fraction of each test cohort, taking fewer cases at an exact half-integer tie.

We used 10,000 patient-bootstrap replicates per route (seed 20261106) to separate three sources of variation: test-cohort sampling at the original cutoffs; development resampling and cutoff recalibration with the original test cohorts unchanged; and both together. Cohorts were sampled independently, with paired A/B scores and labels retained within every record. Repeated draws were retained. Equal-review ranking selected exactly 50 external draw records, ordered by descending risk, original identifier string and draw-position index. The strict original direction pattern required external B-minus-A detection above zero at transferred cutoffs and below zero at equal review; ties were reported separately. Percentile 95 % intervals are descriptive, nominal sensitivity intervals and do not replace the prespecified Newcombe intervals.

We also used symmetric Kitagawa accounting and direct standardization across dataset-reported benign/malignant strata [13]. The external-minus-internal difference was partitioned into a composition-associated term and a within-stratum-rate term. Common weights were the arithmetic mean of the two cohort weights. Detection used the diagnostic shares among low-quality masks, whereas review fractions used shares among all scored patients. We obtained nominal percentile intervals from 5,000 independently sampled cohort-level patient-bootstrap replicates (seed 20261107), recalculating all weights and rates. Undefined stratum denominators remained missing. This accounting describes the observed mixtures and does not estimate causal effects.

Computational reproducibility

The original study assessed computational reproducibility by recomputing statistics from archived case scores and by rerunning all 1,316 images in the original computing environment. Its recorded comparisons covered eight numerical fields, quality labels, review decisions, calibration and review-curve counts. For the present revision, we additionally replayed the six A-missed external cases on a CPU using the checksum-identical original checkpoint, exact SAM source and frozen float32 prompt rules. We saved the initial and four shifted predictions for each case and compared all eight numerical fields and decisions with the archive. The two exercises have separate environments and receipts; all cohort analyses continue to use the original scores.

Results

Cohort yield and development calibration

All 213 development, 851 internal test and 252 external test cases were scored. Low-quality masks occurred in 24/213 (11.27 %) development cases, 93/851 (10.93 %) internal cases and 16/252 (6.35 %) external cases. Each method flagged 43/213 (20.19 %) development cases. The locked thresholds were 0.13705134391784668 for A and 0.06972856653991777 for B.

No initial prediction was empty in the 1,316 study images. All 5,264 initial-to-shifted consistency Dice values were strictly positive, establishing that all shifted predictions were also nonempty. Thus the empty-empty Dice convention did not affect any recorded risk score or review decision.

Performance at the locked thresholds

For H1, method A flagged 71 of 93 low-quality masks in BUS-BRA internal testing (76.34 %) and 10 of 16 in BrEaST (62.50 %; Wilson 95 % CI 38.64 to 81.52 %). The external-minus-internal difference was −13.84 percentage points (95 % CI −38.85 to 7.45). The interval includes zero and is compatible with both a substantial decrease and a smaller increase in detection (Tables 2 and 3 and Figure 3).

[Place Tables 2 and 3 near here]

For H2, the method A simulated review fraction was 174/851 (20.45 %) internally and 35/252 (13.89 %) externally, a difference of −6.56 percentage points (nominal 95 % CI −11.25 to −1.10). Among masks with reference Dice of at least 0.70, A flagged 103/758 (13.59 %) internally and 25/236 (10.59 %) externally. The overall rates of missed low-quality masks with A were 2.59 per 100 internal patients and 2.38 per 100 external patients.

For H3, method B flagged 13/16 (81.25 %; Wilson 95 % CI 56.99 to 93.41 %) low-quality BrEaST masks compared with 10/16 (62.50 %) for A. Ten were flagged by both scores, three by B alone and three by neither. The paired difference was 18.75 percentage points (nominal 95 % CI −4.28 to 40.04). B flagged 33/252 (13.10 %) external cases and missed three low-quality masks. In the internal cohort, B flagged 150/851 (17.63 %) cases, detected 68/93 (73.12 %) low-quality masks and missed 25 (Table 2).

Sensitivity and equal-review comparisons

At reference Dice below 0.50, A detected 52/53 internal and 5/7 external low-quality masks, and B detected 43/53 and 5/7, respectively. At Dice below 0.80, the corresponding counts were 109/183 and 21/50 for A and 95/183 and 21/50 for B. With each alternative definition, A's external detection point estimate was lower than its internal estimate, and both methods had the same external detection count (Tables S2 and S3 and Figure S1).

At 50/252 reviews (19.84 %) per method in BrEaST, A detected 14/16 (87.50 %) and B detected 13/16 (81.25 %) low-quality masks. The paired B-minus-A difference was −6.25 percentage points (nominal 95 % CI −31.61 to 19.57). The order of the two detection point estimates differed between the locked-threshold and equal-review settings (Figure 4 and Table S4). Figure S2 shows the paired detections and misses in both settings.

Reproducibility checks

The archived same-environment raw-image rerun scored all 1,316 cases. The maximum absolute difference was zero for each of the eight numerical fields, and all quality and review decisions matched (Table S5). Statistical replay reproduced the thresholds, primary and supplementary results, and count-level review curves. These checks concern computational reproducibility of the recorded pipeline.

Additional descriptive quality-control measures

The post hoc external quality-control counts were TP=10, FP=25, TN=211 and FN=6 for A, and TP=13, FP=20, TN=216 and FN=3 for B. Specificity was 89.41 % for A and 91.53 % for B; PPV was 28.57 % and 39.39 %, and NPV was 97.24 % and 98.63 %, respectively. The complete counts and Wilson intervals are provided in Tables S6 and S7.

Exploratory diagnostic strata

Internal benign- and malignant-labelled groups contained 577 and 274 patients, including 53 and 40 low-quality masks. A detected 51/53 (96.23 %; Wilson 95 % CI 87.25 to 98.96 %) and 20/40 (50.00 %; 35.20 to 64.80 %), while B detected 41/53 (77.36 %) and 27/40 (67.50 %), respectively. Externally, each diagnostic stratum contained eight low-quality masks: A detected 5/8 in each group, and B detected 7/8 benign-labelled and 6/8 malignant-labelled cases. Review fractions and intervals are reported in Table S8. All stratum counts recombine exactly to the overall results.

External missed-case audit

The complete set of six external low-quality cases missed by A comprised three also missed by B (cases 089, 175 and 238) and three detected by B (024, 105 and 222). In the shared-miss group, mean prompt-consistency Dice was 0.9590, 0.9799 and 0.9942, respectively, despite reference Dice of 0.3191, 0.0293 and 0.6813. Table S9 reports all six cases, their archived scores and flags. This is an outcome-defined descriptive case audit, not an additional estimate of population performance.

Review-policy and calibration sensitivity

Across the complete budget grid, external B-minus-A detection under transferred cutoffs was positive at targets of 5–20 %, negative at 25 %, 30 % and 50 %, and zero at 40 %. Under equal-review ranking, it was positive at 5–15 %, negative at 20 %, and zero at 25–50 % (Table S10 and Figure S3). Neither score consistently detected more low-quality masks across these policies and budgets.

The strict 20 %-target direction pattern occurred in 39.61 % of fixed-cutoff test-sampling replicates, 90.40 % of development-only calibration replicates and 30.94 % of full development-and-test replicates. At least one comparison was tied in 28.08 % of full replicates. Full-resampling percentile intervals were [−39.03, +11.71] percentage points for A detection and [−12.70, −1.55] for its review fraction, in external-minus-internal order (Table S11 and Figure S4). These frequencies describe empirical resampling, not p-values or population probabilities. At the original fixed cutoffs, the absence of an observed A-only low-quality detection structurally constrains the test-sampling bootstrap difference to be nonnegative; its [0.00, +40.00] percentile interval cannot exclude a population A advantage. The original H3 Newcombe interval remains unchanged.

Diagnostic composition and standardized differences

For A, the −6.56-point external-minus-internal review difference comprised a +0.64-point composition-associated term and a −7.20-point within-stratum-rate term. Its detection difference had corresponding terms of −1.62 and −12.23 points. For B, the review terms were +0.79 and −5.32 points, and the detection terms were −0.78 and +8.91 points (Table S12 and Figure S5). The within-stratum term equals the difference after standardization to common diagnostic weights. Detection components were defined in 4,999/5,000 replicates; one replicate had no external benign low-quality case. Overall differences and all review components had 5,000 valid replicates.

Spatial inspection from targeted CPU replay

The six-case CPU replay saved 30 nonempty prediction masks. All 12 A/B review decisions and six quality labels matched the original records. Initial reference Dice differed by at most 9.61×10^{-5}; the largest difference across all 48 numerical comparisons was 0.00143543 for left-shift consistency Dice in case105 (Table S13). Figure 5 shows the three shared misses, with all six cases in Figure S6. In case175, the predicted region lay along the upper image edge and had little overlap with the released reference (replayed Dice 0.0293), while both risks remained below their review cutoffs. These are newly captured CPU masks; the original six CUDA arrays were not archived, so pixel identity could not be tested.

Discussion

Decision rules and deployment workload

A fixed numerical cutoff and a fixed review count make different commitments when a quality-control score is transferred. A cutoff can be applied case by case, leaving the resulting review fraction as an outcome. Ranking fixes the number selected within a defined cohort or batch. In the 20 %-target example, B detected 13/16 low-quality external masks at its transferred cutoff and A detected 10/16; at 50 reviews each, the counts were 13/16 and 14/16. The complete budget scan showed that this ordering depended on the target as well as the rule. Evaluating both policies therefore distinguishes the transfer of a development operating point from detection at a specified review capacity.

Method A selected approximately one fifth of the internal cohort and one seventh of the external cohort for review. Its external detection contrast was less precise: workload was estimated from all 252 external patients, whereas detection used the 16 low-quality masks. The review fraction combines the prevalence of low-quality masks with the flagged fractions in low-quality and acceptable groups. BrEaST had both a smaller low-quality fraction and a smaller flagged fraction among acceptable masks. Misses per 100 patients depend on that prevalence too, explaining why they can remain similar while detection point estimates differ.

The diagnostic-stratum analysis gives a more specific description of this workload difference. For A, the positive composition-associated term and negative within-stratum-rate term act in opposite directions; the latter is also the common-weight standardized difference. Thus the two-stratum accounting locates the observed difference in rates within the diagnostic groups, without identifying its cause. For detection, these weights describe the low-quality subset, not all patients. Dataset differences in acquisition, devices and reference annotation remain combined with the observed diagnostic mixtures.

What prompt disagreement adds

Prompt disagreement added three detections at the locked external operating point. Its yield relative to confidence was not uniform across budgets, and the bootstrap characterized the sensitivity of the 20 % direction pattern to calibration and test sampling. The development-only result is conditional on the original test patients and rankings; the full-resampling result also incorporates their empirical variation. Together with the original paired interval, these analyses support a policy-specific interpretation of the observed counts rather than a general ordering of the scores.

Confidence and prompt disagreement capture different properties of the same initial prediction. A uses the model quality estimate, whereas B probes stability under four specified box translations. The replay contours make their shared blind spot visible: a consistently selected region can remain poorly aligned with the reference. Case175 is an inspectable example of this behavior, with a narrow upper-edge prediction and very low reference Dice. Prior work combined structural responses with confidence [4] and directly compared perturbation uncertainty with predicted IoU [5]. The present evaluation complements those approaches by jointly reporting detection, review volume and misses under explicit operating policies.

Limitations and implications

The findings apply to reference-informed, known-location boxes. Reader-drawn or automatically generated boxes require separate evaluation. The study used one SAM checkpoint, one image per patient and one independent external dataset; overlap with the SA-1B pretraining archive could not be determined [1]. Sixteen external low-quality masks limit detection precision, and empirical resampling does not add independent patients. The new budget, resampling and accounting analyses are post hoc; they define the scope of the operating example rather than confirming a prespecified population-level reversal.

Reference Dice measures agreement with released contours, and the operational cutoff requires separate clinical justification. Simulated review counts do not measure reader time, correction success or downstream measurement accuracy. Future evaluations should predefine the review policy and assess detection and workload jointly with independently supplied prompts and additional external cohorts.

All recorded predictions were nonempty. In future data, simultaneous empty initial and shifted masks could nevertheless receive zero disagreement risk under the stated consistency convention; any explicit empty-mask review rule would be a prospective policy change.

In this known-location breast ultrasound evaluation, transferring development-calibrated numerical cutoffs changed the fraction of cases flagged for review. The relative detection yield depended on the operating policy and budget, with no uniformly better score across the complete grid. Joint reporting of detection, review fraction and missed cases, together with an explicit choice of fixed cutoff or fixed review count, provides a practical basis for interpreting segmentation-quality screening across datasets.

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used the Scientific Agent Skills workflow [14]. A precise underlying Codex model identifier was not available in the archived study records. The human authors retain responsibility for the work.

Research ethics. This retrospective secondary analysis used lawfully obtained, publicly released anonymized breast ultrasound images and annotations and involved no participant recruitment or intervention. The authors confirmed that the study met the conditions for exemption from ethics review under Article 32 of the Chinese Measures for Ethical Review of Life Science and Medical Research Involving Human Participants (2023).

Author contributions. All authors have accepted responsibility for the entire content of this manuscript. Haoze Lan: research conceptualization and principal study design. Ziyuan Zhu: code review, GitHub repository uploads and manuscript revision.

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Informed consent. No participants were recruited or contacted for this secondary analysis of publicly available anonymized data, and no additional informed consent was obtained.

Data and code availability. The original BUS-BRA version 1.0 images and reference masks are available at https://doi.org/10.5281/zenodo.8231412 and BrEaST version 1 at https://doi.org/10.7937/9WKK-Q141 [8, 10]. Study-specific manifests, case scores, scoring and statistical code, and reproducibility records are available in the research repository [15]. The original study archive is retained at commit 592566b4c5f39e261c676eb034139a974e9cdfd5; the additional budget, bootstrap, diagnostic-accounting and six-case replay evidence is archived at commit 6a7da8181db6ba3db0bb9fa8bef628f284791b2e. Original datasets and derived materials are distinguished in the data, experiments and reproducibility directories. The checkpoint is not bundled; documented source URLs and its SHA-256 allow identity verification. The supplementary materials include the new tables, actual replay contours and executable analysis references.

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Tables and table legends

Table 1. Study cohorts

Cohort	Patients, n	Benign, n	Malignant, n	Low-quality masks, n/N (%)
BUS-BRA development	213	145	68	24/213 (11.27 %)
BUS-BRA internal	851	577	274	93/851 (10.93 %)
BrEaST external	252	154	98	16/252 (6.35 %)

One image was selected per patient. Low-quality masks had a reference Dice coefficient below 0.70. n, number of patients; N, cohort size.

Table 2. Test performance at the locked review thresholds

Test cohort	Score	Detection, n/N (%) [Wilson 95 % CI, %]	Simulated review, n/N (%)	Misses, n	Misses per 100 patients
BUS-BRA internal	A	71/93 (76.34 %) [66.77, 83.83]	174/851 (20.45 %)	22	2.59
BUS-BRA internal	B	68/93 (73.12 %) [63.32, 81.08]	150/851 (17.63 %)	25	2.94
BrEaST external	A	10/16 (62.50 %) [38.64, 81.52]	35/252 (13.89 %)	6	2.38
BrEaST external	B	13/16 (81.25 %) [56.99, 93.41]	33/252 (13.10 %)	3	1.19

Detection uses low-quality masks as its denominator. Simulated review and misses per 100 use all scored patients in each cohort. A, model-confidence score; B, prompt-disagreement score; n, number of events; N, relevant denominator. Brackets show Wilson 95 % intervals for detection, in percent.

Table 3. Planned contrasts at the locked review thresholds

Analysis	Contrast	Difference, pp	95 % CI, pp
H1 (primary)	Method A detection, external minus internal	−13.84	[−38.85, +7.45]
H2 (secondary)	Method A simulated review, external minus internal	−6.56	[−11.25, −1.10]
H3 (secondary)	External detection, Method B minus Method A	+18.75	[−4.28, +40.04]

pp, percentage points. CI, confidence interval. H1 and H2 use independent Newcombe intervals. H3 uses a paired interval in the 16 low-quality BrEaST cases. H2 and H3 intervals are nominal.

Figure legends

Figure 1. Study cohorts and analysis flow. BUS-BRA patients were assigned to development or internal testing. BrEaST contributed external test patients with a primary lesion mask. Each patient contributed one image. Low-quality masks had a reference Dice coefficient below 0.70.

Figure 2. Review scores for prompted breast ultrasound segmentation. (A) The initial box is derived from the manual contour with a 10 % margin per side and clipped to the image. Four further prompts shift the clipped box by 5 % of its width or height. (B) Frozen Segment Anything Model (SAM) ViT-B reuses one image embedding to generate one mask per prompt. (C) Method A uses one minus predicted intersection over union for the initial mask. Method B uses one minus the mean Dice between the initial and four shifted masks. Each score triggers simulated review at or above its locked cutoff. The image and masks share a display crop from BrEaST case 001 [10].

Figure 3. Planned differences at the locked thresholds. Points show differences in proportions and horizontal bars show two-sided 95 % Newcombe confidence intervals. H1 and H2 compare BrEaST minus BUS-BRA internal testing. H3 compares B minus A in the same 16 low-quality BrEaST cases. H2 and H3 intervals are nominal. The dashed line marks zero difference.

Figure 4. Low-quality mask detection by simulated review volume. Panels A and B show the percentage of low-quality masks detected as progressively more patients are selected in descending risk order in each test cohort. Markers identify the locked thresholds and the equal-review setting. Panel C compares detection in the same 16 low-quality external cases at the locked thresholds and at 50 reviews per method.

Figure 5. Three shared screening misses with actual targeted CPU predictions. Rows show BrEaST case089 (A, B), case175 (C, D) and case238 (E, F) [10]. Left panels overlay the released reference contour (green) and reference-derived initial prompt (blue); right panels overlay the initial prediction captured in the CPU replay (orange). Both review scores missed these three low-quality masks at the original locked cutoffs. Predictions use the original checksum-matched SAM ViT-B weights, exact source and prompt rules. Displayed Dice values come from this replay. The maximum initial-reference-Dice difference across all six replayed cases was $9.61 \times 10{-}5$; original study scores remain in Table S9. All six replayed cases and numerical comparisons appear in Figure S6 and Table S13.